Measuring Transit Accessibility in Allegheny County

User guide for GIS dashboard and helper tools

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Table of Contents

Introduction 5

Getting Started 6

- Quickstart Guide 6
- System Requirements 6
- Expected Audience and Use 6

What's in the Transit Access and Scripts Zipped Folder? 7

- GTFS Folder 7
- Capstone_CSVs Folder 7
- Census Analysis Folder 7
- Opportunity Zone Analysis Folder 8
- ZoneData 8
- APC-AVL Folder 8
- Google API Folder 9

Identifying Regions of Interest ("Origins") 9

- U.S. Census Bureau data 9
- DHS Client Data 10

Identifying Regions of Interest ("Destinations") 11

Building the Opportunity Employment Accessibility Index 12

- Components of the Index 12
- Extracting Transit Data from the Transit Tool 12
  - Identifying Your Bus Stops of Interest 13
    - Manual Selection 13
    - ArcGIS as a Subsetting Tool 13
  - Integrating GTFS CSVs into the Index 13
- Extracting Car Travel Times 14
- Extracting Walkability Data 14
- Extracting Geographic Coverage 14
- Extracting Job Counts 15
- Extracting Relevant Bus Stops 15
- Computing Key Metrics 15

Bus Travel Time 15
Cost 15
Reliability 16
Bus to Car Alternative 16
Walkability 16
Geographic Coverage 16
Normalizing the Intermediate Scores (e.g. Travel Time, Walkability) 16
Time-Window Score 17
Composite Score 17
Changing Key Metric Weights 17
Understanding the Excel Index Columns and Formulas 18
Index Interpretation 20
Interpreting the Index: Within a Time Window 20
Interpreting the Index: Across Time Windows 20
GIS Dashboard 21
FAQs 25

Introduction

Allegheny County is the second largest county by population in Pennsylvania. With over one million people living in the county (U.S. Census Bureau), reasonable and reliable means of transportation are essential. The Allegheny County Department of Human Services (DHS) is interested in measuring how the Port Authority of Allegheny County's (PAAC) transit system impacts access to employment. This led to the research question:

How accessible are opportunity occupations in Allegheny County by public transit?

This user guide provides context for interpreting the opportunity accessibility index and associated visualizations. The user guide also provides instructions on how the index can be modified and reweighted for the purposes of DHS redeploying on specific projects within the department.

Getting Started

Quickstart Guide

To login to the GIS dashboard users will open the link DHS Opportunity Zone Analysis Dashboard.

Additionally, in the supplementary files provided, there are R scripts to be run at separate parts of the process of generating an index.

System Requirements

- Microsoft Excel 2016 MSO (16.0.12228.20100) 32-bit Version 1911
- ArcGIS Pro 2.4.1
- R Version 3.6.1/ RStudio Version 1.2.1335 (recommended)
    - R Version 3.5 + should work fine
- Internet access to connect to PAAC API for R tool

Expected Audience and Use

The ArcGIS dashboard was designed with DHS employees in mind. From this tool DHS Employees, in an integrated workspace, will be able to:

- Identify gaps in infrastructure possibly preventing clients from finding gainful employment
- Suggest service lines in need of review to PAAC and recommend solutions (e.g. first-last mile solutions) where appropriate

What's in the Transit Access and Scripts Zipped Folder?

Below is a map all of the files and folders you will receive from the transit folder.

GTFS Folder

Name	Type	Description
BG Mapping, Origin Mapping, Destination Mapping	Excel	These were used to simplify the mapping process for our custom polygons and block groups in the capstone project. The PGH Census Tracts file is the inclusive list of all block groups and neighborhoods in Allegheny County.
Loop Free Raptor	R Markdown	The raw script to generate CSV files for each time of day based on a provided list of origins and destinations
PGH Census Tracts	Excel	Excel sheet with list of all neighborhood names associated with block groups (WPRDC)
Transit Station Info	Excel	Contains demographic information for transit stations
app	R-Shiny Dashboard	Executable script for DHS transit analysis helper tool.

Capstone_CSVs Folder

Contains the csv output from running the untampered Loop Free Raptor given the origins and destinations selected for the capstone project.

Census Analysis Folder

Name	Type	Description
census_data_cleaning	R script	R scripts to clean and pre-process ACS data for further origin and destination analysis

Opportunity Zone Analysis Folder

This folder was used to generate top employers and NAICS code, and includes the raw ReferenceUSA datafile.

Name	Type	Description
Industry Match	Excel	Maps NAICS industry codes with industry description
ZoneData	R Markdown	Processes raw Business USA file to analyze top industry and employer given block group input
Zones_AndJobs_TableToExcel	Excel	List of all employers/business data within our identified opportunity zones
Reference USA Data_Top20OppsIndustries.csv	CSV	List of all employers/business data in 20 identified opportunity industries with 10+ employees, as of late October 2019

APC-AVL Folder

Name	Type	Description
1703	Excel	Sample raw repository of Pittsburgh Data - APC helper (March 2017)
Google Transit Clever Trip	Zip file	Demographic information regarding bus routes, schedules, trips and stations.
APC_AVL	R file	Commented R script file detailing steps for APC only or APC-GTFS case studies

Google API Folder

Name	Type	Description
Car Travel Time Module	R file	R script containing Google Maps API access commands
Stop2StopTravelTimes	CSV file	Sample output matrix of car travel times from google API

Identifying Regions of Interest ("Origins")

This section will detail the process followed to collect and analyze the data that were used for selecting areas of interest, otherwise called "origins". Both ACS and DHS proprietary data on clients were used to define regions of interest.

U.S. Census Bureau data

1. Download U.S. Census Bureau block group-level demographic and socioeconomic data (ACS 5-year 2013-2017; comma-separated value format) from American FactFinder on:
    a. Population
        i. B01001: Sex by Age
        ii. https://factfinder.census.gov/bkmk/table/1.0/en/ACS/17_5YR/B01001/0500000US42003.15000
    b. Poverty
        i. B17017: Poverty Status In The Past 12 Months By Household Type By Age Of Householder
        ii. https://factfinder.census.gov/bkmk/table/1.0/en/ACS/17_5YR/B17017/0500000US42003.15000
    c. Educational Attainment
        i. B15003: Educational Attainment For The Population 25 Years And Over
        ii. https://factfinder.census.gov/bkmk/table/1.0/en/ACS/17_5YR/B15003/0500000US42003.15000
    d. Households headed by single parents
        i. B17010: Poverty Status In The Past 12 Months Of Families By Family Type By Presence Of Related Children Under 18 Years By Age Of Related Children
        ii. https://factfinder.census.gov/bkmk/table/1.0/en/ACS/17_5YR/B17010/0500000US42003.15000
    e. Car ownership
        i. B25044: Tenure by Vehicles Available

ii. https://factfinder.census.gov/bkmk/table/1.0/en/ACS/17_5YR/B25044/0500000US42003.15000

2. Analyze census data (see Census Analysis/census_data_cleaning script) to identify block groups with:
   a. high number of persons aged 18-44
   b. high levels of poverty
   c. high number of households headed by single mothers
   d. low levels of car ownership
   3. Combine the datasets into a single master CSV file

4. Download U.S. Census Bureau GIS data on Allegheny County block groups from the Allegheny County GIS data portal:
   a. ArcGIS file geodatabase (.gdb): https://openac-alcogis.opendata.arcgis.com/datasets/2c059d0501e5479ca90242a6d716d24c_0
   5. Merge TIGER shapefiles with census data in ArcGIS Pro
   a. Add Allegheny_County_Block_Groups_2016 File Geodatabase Feature Class to ArcGIS Pro map
   b. Add combined census data CSV to ArcGIS Pro map
   c. Use GEOID field in file geodatabase to match to GEO.id2 field in census CSV
   6. Map and visualize merged block group data using ArcGIS Pro

DHS Client Data

To map proprietary, anonymized client (DHS) data, including residence and employment information, and to identify areas where many DHS clients live and work the following steps are followed:

Simple processing of data for ArcGIS:

1. Data provided by DHS has location data (at the census block level) of where clients reported they lived and worked (if applicable).
   a. Cleaning: it may be desirable to clean data and remove any observations that did not report the location of their home address (home "block")
   2. Get counts of total number of clients living in a block / block group
   a. Our group opted to aggregate to the block group level
      i. Using data processing software of choice (R, Python, Excel), eliminate the last four digits of the 16 digit block code (block groups are defined by the first 12 digits)
   b. Using data processing software of choice, get total counts for the number of clients living in each block group
   3. Join data in ArcGIS to Allegheny county block layer
   a. Our group opted to aggregate to the block group level in ArcGIS
      i. Allegheny County block groups may be found here: https://data.wprdc.org/dataset/allegheny-county-census-block-groups-2016

Identifying Regions of Interest ("Destinations")

In building out this tool, the team used the following criteria to determine the specific destinations of interest:

1. Download and filter ACS PUMS data:
    a. Restrict respondents to those aged between 16 and 40 (to capture working age population without conflating wages with natural gains due to years of experience)
    b. Remove individuals enrolled in school
    c. Observe the total number of jobs in the dataset associated with a specific industry.
2. Identify opportunity occupations by the following criteria:
    a. Isolated individuals who worked 50-52 weeks in the year, and for each of these weeks worked 35-60 hours per week.
    b. Respondents with less than a bachelor's degree but made more than the median annual wage in Pennsylvania (at the time of the study, this is roughly $59,195).
    3. To identify opportunity occupation zones:
    a. Compute the number of opportunity jobs associated with an industry.
    b. Create an industry opportunity share for each of the observed industries in the NAICS category.
    c. Download every registered business in Allegheny County fitting the following criteria from ReferenceUSA database:
    d. Under Number of Employees, select all except 1-4 and 5-9
    e. Limit industry codes to "Opportunity Industries" 11, 21, 22, 23, 31, 32, 33, 42, 44, 45, 48, 49, 51, 52, 53, 54, 55, 56, 61, 62, 71, 72, 81,92
    f. Estimate number of opportunity jobs at each business based on previously calculated industry share (multiply the total number of employees at each business times industry opportunity share to get an estimate of the number of jobs which may be "opportunity jobs.")

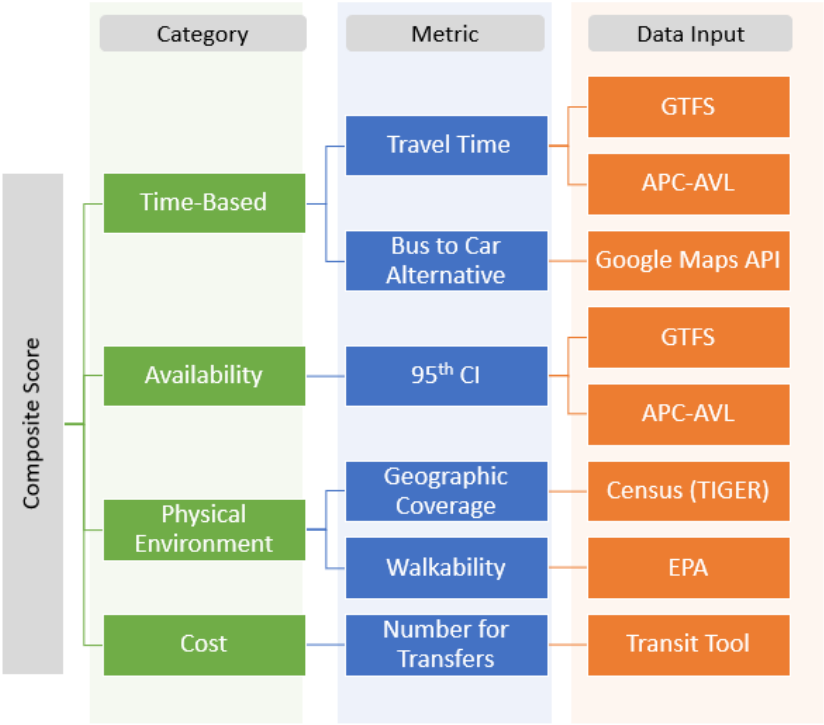
Next, the selection of the destinations was performed in ArcGIS by following the steps below:

1. Plot all businesses by latitude and longitude on to map of Allegheny County using XY to Point analysis tool.
2. Create kernel density Raster map based on the number of opportunity employment jobs estimated to be at each business address (based on the business's industry code) to uncover job hotspots.
    a. Note that for the opportunity job destination zones, this was done in two stages, once for Allegheny County excluding the City of Pittsburgh and then for only the City of Pittsburgh. This was to identify geographically diverse zones; analyzing the whole county at once focused most of the density on downtown, which is accurate but not as interesting as a test case for the transit index.
3. Convert hotspots to polygons using Contour tool
4. Create .25 mile buffer around each zone to capture relevant bus stops

Building the Opportunity Employment Accessibility Index

Components of the Index

The Opportunity Employment Accessibility Index is composed of six distinct indicators pulled from unique sources. The below diagram is helpful to understand where each of the indicators come from:



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Extracting Transit Data from the Transit Tool

Method: R Scripts (see GTFS/app.R & APC_AVL/APC_AVL.R)
Source: GTFS, APC-AVL
Covers: *Average Travel Time, Average Transfers, and 95th Confidence Interval*
The main unit of analysis for the transit tool is bus stops. For GTFS scheduled data, it is necessary to identify the names of bus stops (Internal Stop ID) to feed into the tool. For APC-AVL data, the Latitude and Longitude are needed for this analysis. This data collection represents the trickiest part of the analysis, as the scripts we have provided in the folders take care of much of the rest of the analysis.

R Code Inputs:

- Origin_Stop_List
- Destination_Stop_List
- GTFS feed from PAAC API
- APC_AVL data from PAAC

Identifying Your Bus Stops of Interest

These bus stop locations can be identified in a number of ways:

Manual Selection

If you have the name or PAAC stopID of a bus stop, you can simply import them into the tool as the bus_stop_origins csv file.

ArcGIS as a Subsetting Tool

To calculate travel time, it is helpful to have a subset list of relevant bus stops instead of all 7,001 in Allegheny County. To subset bus stops based on their locations, use the "Intersect" tool in ArcGIS Pro with the following parameters:

- Input Features:
  - PAAC Stops
  - Locations of interest polygon layer
- Output Feature Class: what you would like the subsetting list of bus stops to be called

To export the resulting feature class to a spreadsheet, use the Table to Excel tool in ArcGIS Pro.

Integrating GTFS CSVs into the Index

The Index spreadsheet will have a tab called "WideWindowMatch"

Requirements: Unique identifiers for your origins and destinations, populated into a VLOOKUP table as well as the index itself.

1. Sort the unique pairing ID in your target index worksheet and in WideWindow match alphabetically/ordinally.
2. Copy and paste selected origins and destinations from the CSV into the origins and destinations columns on the ***left hand*** side (**NOTE:** Do not alter the concatenate column)
3. Copy and paste average travel time, average transfers, and CI interval information on the ***right hand*** side of the concatenate column in the WideWindowMatch sheet..
   a. Clear any leftover data from the selected columns
4. Vlookup table on the right side of the sheet shall automatically populate metrics for all pairs based on the left-hand side input
5. Paste WideWindow values into the car travel time column in the index (with more time, we would have created a less manual process, but this works fine!)

Extracting Car Travel Times

Method: Google API queries
Car travel times were extracted using the Google API and the R helper script (see Google API folder - Car Travel Time Module.R).

1. Using the selected origins and destinations Google's API was queried to return all combination of O-D pair travel times across different time windows.
2. The prior step was iterated for each set of queries to:
    a. Ensure time of day is within relevant time window
    b. Calculate travel times from each origin to each destination
    c. Use the Car Travel Match sheet in the Index document to index match travel times
    d. Import car travel values into the relevant column in the index

Extracting Walkability Data

Method: Web Download
Source: [EPA Walkability Index]
Walkability scores for all block groups in Allegheny County have been provided in the Accessibility folder. However, they can also be downloaded here: https://www.epa.gov/smartgrowth/smart-location-mapping#walkability website.

Average walkability scores for areas which are not defined by block groups (e.g. destination polygons) are calculated using "Summarize Within" in ArcGIS Pro and the following parameters:

- Input Polygon: layer including geographic areas of interest (in our case, Block Group origins and Destination Zone polygon destinations)
- Input Summary Features: EPA Walkshed layer
- Summary Fields:
    - Field: data to summarize, in our case, D3b Ranked
    - Statistic: mean

To export the resulting feature class to a spreadsheet, use the Table to Excel tool in ArcGIS Pro.

Extracting Geographic Coverage

Method: ArcGIS
To approximate how much of the geographic area is within the Port Authority's Walkshed use the "Summarize Within" tool in ArcGIS Pro and the following parameters:

- Input Polygon: layer including geographic areas of interest (in our case, Block Group origins and Destination Zone polygon destinations)
- Input Summary Features: PAAC Walkshed layer
- Shape Unit: Square Kilometers as the Shape Unit.

To export the resulting feature class to a spreadsheet, use the Table to Excel tool in ArcGIS Pro.

Extracting Job Counts

Method: ArcGIS

To estimate how many jobs are accessible within the walksheds of destination zones, use "Summarize Within" in ArcGIS Pro with the following parameters:

- Input Polygon: layer including geographic areas of interest (in our case, Block Group origins and Destination Zone polygon destinations)
- Input Summary Features: BusinessLocations layer
- Summary Fields:
  - Field: data to summarize, in our case, "Number of Opportunity Jobs"
  - Statistic: sum

To export the resulting feature class to a spreadsheet, use the Table to Excel tool in ArcGIS Pro.

Extracting Relevant Bus Stops

Method: ArcGIS

To calculate travel time, it is helpful to have a subset list of relevant bus stops instead of all 7,001 in Allegheny County. To subset bus stops based on their locations, use the "Intersect" tool in ArcGIS Pro with the following parameters:

- Input Features:
  - PAAC Stops
  - Locations of interest polygon layer
- Output Feature Class: what you would like the subsetting list of bus stops to be called

To export the resulting feature class to a spreadsheet, use the Table to Excel tool in ArcGIS Pro.

Computing Key Metrics

Bus Travel Time

1. In the GTFS folder open and run the app.R file.
2. When prompted select the appropriate directory containing the GTFS data files (neighborhood mapping files and selected origin and destination files).
3. On the input pane on the left hand side of the app, select the desired parameters and click submit.
4. Once chart and summary table are populated export summary data using the "export to csv" button on the input selection pane.
5. Import the csv from step 4 into the index spreadsheet.

Once these are imported from the transit tool, minimal processing is needed.

Cost

This is a function of total transfers. The formula is

- Costs = $2.75 + (# of Transfers * $2.75)
- $2.75 represents the current cost of fare in Allegheny County

Reliability

While the confidence interval alone communicates key information, reliability is measured as the distance between the average travel time and the 95th CI. This indicates how much variation (or spread) there is in travel time by capturing "how bad it could get."

Variation = absolute value (confidence interval - average_travel_time)

Bus to Car Alternative

The Bus Car Ratio is the average bus travel time divided by the average car travel time

- Bus/Car Ratio = Bus Travel Time / Car Travel Time

Note: Car times will generally be shorter.

Walkability

The walkability metric is based on a simple average of the walkability surrounding a particular origin departure and the destination's arrival.

- Formula = (Origin Walkability + Destination Walkability) / 2

Geographic Coverage

This portion represents what percent of an origin's physical area is within the Port Authority Walkshed. This is usually fed in directly from ArcGIS summary tools.

- Formula = Walkshed Geographic Area Within Buffer / Total Geographic Area

Normalizing the Intermediate Scores (e.g. Travel Time, Walkability)

What's going on in this section?

Min Value	0.1	0.02	0.0	0.0	0.0	0.0
Max Value	1.2	1.15	1.6	1.4	1.5	1.5
Span	1.1	1.13	1.6	1.4	1.5	1.5

	Done	Done	Done	Done	Done	Done	Done
s	Logistics	7.5	15.0	15.0	15.0	15.0	15.0
	float	Score	Score				
alk	Avg. Walk	Coverage_Score	ability_morning_peak_Sc	liability_day_offpeak_Sco	iability_evening_rush_Sc	ibility_evening_offpeak_S	reliability_12a-6p_Score
0.98	1.0	4.72	11.8	10.4	12.6	9.7	15.0
0.31	0.7	4.72	15.0	14.9	15.0	14.9	11.8
0.86	0.9	5.48	10.1	9.4	12.0	13.1	0.0
0.88	0.9	4.01	14.7	13.7	13.8	14.5	0.0

The method for normalization here is relative to all other pairs. For each intermediate column

1. Identify the maximum and minimum values
2. Divide each observation by the span (max - min)

NOTE: This is a simple scoring method; future work could incorporate percentile methodologies, although the simple scoring method is pretty common across the transit scoring space.

- Value = Raw Data Input
- Static Factors Intermediate Score -- *Big is Good*

- Geographic Coverage, Walkability
  - Score = (Observed Value - Minimum) / (Max - Min) * Weight
- Dynamic Factors Intermediate Score -- *Small is Good*
  - **Use For**: # of Transfers, Travel Time, 95th CI, Travel Cost
  - Score = 1- (Observed Value-Minimum)/Max-Min) * Weight

Time-Window Score

This is strictly the sum across <u>all weighted intermediate</u> scores

Composite Score

The average across all time-window scores is the composite score.

Changing Key Metric Weights

Users can adjust weights by:

1. Navigate to the scoring methodology section in the index spreadsheet
2. Change the weights in the orange box

NOTE: Ensure that the sum of weights adds up to 100.

Understanding the Excel Index Columns and Formulas

What's going on in all these columns?

Below are the columns found in the index. The template has many of these features already built in, so that intermediate calculations, intermediate scores, and time-window scores should automatically repopulate with the changes to the raw inputs. Information for extracting and generating that raw input is above.

Legend

Information: Information used to identify origins and destinations
Raw Input: Derived from CSV our Excel output from transit tool Google API, or ArcGIS exports
Calculated Column /Intermediate Result: Necessary transformations to get intermediate scores
Intermediate Score Column: Score for one of the six intermediate metrics
Time Window Score: Overall score for a given time of day
Composite Score: Average score across all time windows

Column Descriptions

Origin_BG: the block group used to map to ArcGIS files
Dest_ID: for us, the identifier of out custom opportunity zone polygon
Pairing ID: Custom pair for your origins and destinations. Can be created by just numbering unique combinations
Origin : Origin Name (usually neighborhood label)
Destination: Destination name (usually neighborhood label)
Average Travel Time: CSV input from travel tool (copy/paste). A matching system is included in the WideWindowMatch sheet
Car Travel Time: CSV input from Google API (copy/paste). A matching system is included in the CarTravelTime Sheet
Bus Car Ratio: Bus Travel Time / Car Travel Time (given a window)
Reliability: CSV input from travel tool (copy/paste). A matching system is included in the WideWindowMatch sheet
Variation (Var) : distance between the 95th CI and the average = 95th CI - Average Travel Time
Transfers: CSV input from travel tool (copy/paste)
Cost: Represents costs without a ConnectCard = 2.75 + (2.75 * Number of Transfers)
Area Coverage: Input from ArcGIS (provided). Describes the percent of the zone within the Port Authority Walkshed
Origin_Walk: Input from ArcGIS (provided). Information of the origin's walkability
Dest_Walk: Input from ArcGIS (provided). Information of the destination's walkability
Avg_Walk: Calculated column = average (Origin_Walk, Dest_Walk)
Coverage_Score: Generated from weights and correct max and min
Availability_Score: Generated from weights and correct max and min
Walkability Score: Generated from weights and correct max and min
Cost_Score: Generated from weights and correct max and min

Travel_Time_Score: Generated from weights and correct max and min
Bus_Car_Ratio_Score: Generated from weights and correct max and min
Time_Window_Scores: Time Window composite based off weights = sum(Intermediate Scores)
Composite_Score: Average of all time window scores = average (all_time_scores)

Index Interpretation

Interpreting the Index: Within a Time Window

An Origin-Destination Pair

	Cost	Average Travel Time	Bus-Car Alt	Expected Travel Time Range	Walkability	Geographic Coverage
Raw Data	$4.68	1.36 hours	3.6	0.5 hours	14/20	30%
Processing	*Normalization*					
Possible Points	30.0	20.0	20.0	15.0	7.5	7.5
Total	21.1	7.5	12.5	8.3	4.4	5.3

An interpretation of the above image is as follows:

"The average cost from this origin to this destination is about **$4.68**. The average trip time is **1.36**
Hours (~1 hour, 25 min**)**, which is **3.6** times longer than a car ride would be. I can expect to wait at most an additional **30 minutes** (0.5 hours) above this expected travel time. The EPA Walkability score is **14/20** (barely higher than average), and **30%** of this zone is within the Port Authority Walkshed."

Interpreting the Index: Across Time Windows

Reading across the index is relatively straightforward. If a route exists within the time window, the score will be populated in the column. If not, the score will be 0.0 and the box will be red, indicated inaccessibility.

Morning_Peak_Score	Day_Offpeak_Score	Evening_Rush_Score	Eve_Offpeak_Score	Night_Score	Weekend_Score
82.40	79.60	79.56	75.90	0.00	72.66

An interpretation of the above image is as follows:

"This route is relatively accessible during weekends and daylight hours, but is not accessible at night between 12am and 6am."

GIS Dashboard

Method: ArcGIS Pro and Operations Dashboard

Account information for using dashboard:
Gmail Account

- Email: DHS.CMU.Systems.Dashboard.F19@gmail.com
- Password: AlleghenyCounty2019
- First name: DHS
- Last name: Human Services
- Recovery question: What city were you born?
- Answer: Pittsburgh
- Birthdate: January 1, 1990

ESRI Account - trial

- Username: DHS_Systems_Dashboard_F19
- Password: AlleghenyCounty2019
- Email: DHS.CMU.Systems.Dashboard.F19@gmail.com
- Recovery question: What city were you born?
- Answer: Pittsburgh
- Phone number: (555) 555-5555

Maps Imported into the Dashboard Map
See previous instructions above on how the following shapefiles were created in GIS and then add to dashboard map:

1. Opportunity Employment Zone Centroids (make sure this includes XY coordinates)
    a. quarter-mile buffer
2. Origin Block Groups
    a. quarter-mile buffer

Add the following layers:

1. Allegheny County boundary
    a. Source: http://openac-alcogis.opendata.arcgis.com/datasets/allegheny-county-boundary
2. Allegheny County municipality boundaries
    a. Source: http://openac-alcogis.opendata.arcgis.com/datasets/allegheny-county-municipal-boundaries
3. PAAC routes (line feature)
    a. Source: https://data.wprdc.org/dataset/port-authority-current-transit-routes - Nov 2018 - March 2019
4. PAAC stops
    a. Source: https://data.wprdc.org/dataset/port-authority-of-allegheny-county-transit-stops
5. PAAC Walkshed

        a. Source: Port Authority of Allegheny County
    6. Allegheny County Water Features
        a. Source: https://data.wprdc.org/dataset/allegheny-county-hydrology-areas

Formatting the Data for the Dashboard

Opportunity Occupation Zones

1. Create csv of the opportunity occupation zones, the neighborhood most closely associated with the zone, and the corresponding indexID number.
2. Import the opportunity occupation zones csv from step 1 into GIS.
3. Join opportunity zone polygon feature to the csv based on the indexID number in order to carry forward the neighborhood name (to appropriately visualize and label in the dashboard).
4. Import additional origin information via csv that includes:
    a. Population
    b. Households in poverty
    c. Persons with a high school education or above
    d. Households headed by a single mother
    e. Car ownership rates
5. Join the additional information table to the origin table via block group
    a. Create a separate polygon feature of each origin by using select by attribute and named each table the neighborhood associated with the block group
    b. Neighborhood names originated from University of Pittsburgh Library Census guide: https://pitt.libguides.com/pghcensus

Origin-Destination Pairs

1. Generate a near table for each separate origin polygon:
    a. Use the opportunity occupation zones as the near feature
    b. Uncheck the "find nearest" and checked "XX" and enter 10,000 for distance
        i. The generate near table now shows the origin-destination pair between the selected origin and each destination
    c. Join each generate near table to the opportunity employment zone feature to include the employment information in the export
    d. Export each generate near table as a csv
    e. Append block group, municipality, and neighborhood dimensions to each csv
    f. Add a "selector" number (i.e., unique ID) to each origin to use in the visualization tools in ArcGIS Operations Dashboard
    g. Add an objectID column to give each row a unique value to join on in the future
    h. Copy and paste each cleaned csv into one document to be matched to the Accessibility Index based on the origin block group and the opportunity occupation zone
    i. Import the csv back into GIS

Creating the Origin-Destination Pair Lines

1. Use the XY to line function using the “from” and “near” XY coordinates
2. Rejoin the accessibility index to the XY to line feature using the “unique values”
3. Clean the attribute table by reformatting titles of columns and correcting and neighborhood names
4. Update the symbology for the origins, destinations, and line features for best appearance on dashboard.
5. Convert various layers to same projection: WGS 1984 Web Mercator Auxiliary Sphere
6. Clean up file geodatabase to include only layers needed

Creating the Dashboard

1. Upload completed map (created as described above) to ArcGIS online profile
2. To Create new Dashboard:
    a. Click apps tab in the top right corner, new operations dashboard
    b. Add map to dashboard using plus sign
        i. Turn the following tabs “on”:
            1. Pop-ups
            2. Scalebar - ruler
            3. Legend
            4. Layer visibility
            5. Search
            6. Zoom in/out
        ii. Additions to dashboard (plus sign):
            1. Add map to dashboard
            2. Add sidebar
            3. Add selector
                a. Selector process for selecting origin and see the OD pairs:
                1. Select “group values” from categories form
                2. Select O_D_Connections (line feature name)
                3. Select “selector” for category field
                4. Edit the labels for each category selector to be the origin neighborhood
                5. Increase maximum categories to a very large number so as to not accidentally exclude any lines
                6. Select “single” in selection
                7. Select “equal” in operator
                8. Select “dropdown” for display type
                9. Select “first” for default selection
                10. In the “action” tab of the tool:
                    a. Add target to the filter of the same line feature (O_D_Connections)
                11. Using the add button:
                    a. Add header

        i. Customize symbology/font/text, add icon
    b. Add legend
    c. Add rich text (the “About the Dashboard” section)
        i. Add description on how to use the dashboard
    d. Add bar graph
        i. Click add serial chart
        ii. Select Origin to Opportunity Employment Zones layer as the data source
        iii. Select “fields” under categories from
        iv. Select Total Opportunity Jobs and Total Opportunity Jobs within Walkshed
        v. Select average as the statistic
        vi. Change the font colors to white and the bar colors to orange and blue
        vii. Add graph and axis titles
    e. Add gauge/measurements
        i. Add summary statistics modules for” at a glance” takeaways
        ii. Communicate maximum, minimum, and average scores from the accessibility index, along with explanatory characteristics associated with those scores
c. Reformat the dashboard by dragging, dropping, and resizing the different elements for the best appearance
d. Reformat pop-ups in ArcGIS Online WebMap to show only relevant data and titles in dashboard
e. Change permissions in “content” main page to “allow everyone/public” to allow for public access

FAQs

I'm having trouble importing my excel files into the transit helper app?
Excel .csv files to be encoded in UTF-8 format. Double check the file formats of the files intended to upload. I would recommend using Notepad++ and checking the encoding menu item.

I really care about walkability. How do I emphasize this in my findings?
One can adjust weights by going into the scoring methodology section and manually changing the weights in the orange box.

Why Am I Seeing Such Big Numbers for Inaccessible Places?
Currently, the index cannot deal with NAs using the min/max span method. Because of this, locations that are not accessible are flagged in two ways:

1. Time-based columns are automatically populated with MAX values (highest observed value plus a small amount)
2. The transfer column defaults to 4 transfers and a red hue, indicating inaccessibility

These methods guarantee no points are received in these categories, without introducing zeroes and NAs that would otherwise break the calculations.

NOTE: While a number of ifelse functions stop these paris from showing up anywhere in the intermediate and time-based scores, there is an opportunity to clean this process a bit.

Does the dashboard automatically update?
No. All information must be manually inputted into an ArcGIS map, upload to the online account associated with the dashboard, and uploaded to the dashboard. Then, all auxiliary items (such as data visualization or the selector tool) must be updated as well.